



Efficient cooperative localization method with node selection based on position error bound

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ABSTRACT

Cooperative Localization (CL) has become a promising solution that circumvents the needs for high-density anchor deployment, and offers additional localization accuracy in Wireless Sensor Networks (WSNs). However, it suffers from high computational complexity and increased network traffic problem. To alleviate this problem, we propose an iterative greedy node selection approach based on Position Error Bound (PEB) for Angle-of-arrival (AOA) based CL. We introduce a Boolean vector to formulate the node selection optimization problem that maximizes the Decrease of PEB (DPEB) from agents' cooperation. Since the problem is NP-hard and difficult to solve, the iterative greedy algorithm is proposed to approach the sub-optimal solutions. Then, in order to improve the accuracy of localization phase, we derive the efficient closed-form solution of AOA CL problem and propose an iterative constrained Least Squares (LS) method. Moreover, the theoretical analysis shows that the localization estimator is very close to the Cramer Rao Lower Bound (CRLB) performance. Finally, simulation results show that the proposed node selection approach can greatly decrease the number of agents participating in the CL without significant degradation in localization accuracy. Besides, the proposed CL method outperforms the existing ones in terms of localization accuracy.

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1. Introduction

The availability of absolute or relative position information has gained considerable attention over the past decade due to the remarkable development of location-based services. Generally, the location-aware networks consist of two kinds of nodes: anchors and agents, where anchors can obtain their approximate locations (for example, through Global Navigation Satellite System (GNSS) Real-time Kinematic (RTK) or system design), whereas the locations of agents are desired to be estimated. Each agent can localize itself by collecting and processing the signals conveyed from anchors by different measurements, such as Time-of-arrival (TOA), Received Signal Strength (RSS), Angle-of-arrival (AOA), Time-difference-of-arrival (TDOA), or hybrid method [1]. In dense obstacle environment, the existence of multi-path effect, wall sheltering, and channel interference generally causes the significant variations of measurements, and consequently results in the decrease of localization accuracy. Thus, the single agent localization is limited to the number of accessible anchors.

To tackle the problem, cooperative multiple agents localization can improve accuracy and robustness, where it not only utilizes the measurements between the anchors and agents, but also those between the agents. The additional measurements resulted from the cooperation of agents can improve localization performance by enabling the agents to help each other in estimating their locations [2]. In the Cooperative Localization (CL) network, agents attempt to estimate their locations using signal measurements through a centralized or distributed localization algorithm [3]. In centralized algorithm, the measurements from anchors as well as agents are shared with a center processor to compute the agents' locations. In distributed algorithm, every agent locates itself based only on locally collected information. Centralized algorithm utilizes information about the entire network and has better localization accuracy in harsh environment. In this paper, we focus on a centralized AOA based CL network.

Theoretically, the agent can achieve optimal localization performance by cooperating with all other agents within communication range. However, it will result in high energy consumption problem, especially in some resource-constrained network. In order to balance localization accuracy and energy consumption, node selection approach and localization method are critical for CL network. In this paper, we propose an iterative greedy node selection approach

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based on Position Error Bound (PEB, i.e., the square root of trace of the Cramer Rao Lower Bound (CRLB) matrix), and an efficient AOA based CL method. The key contributions of this paper are as follows.

- Considering the correlation of Cooperative Agents (CAs), we derive the PEB of AOA based CL to characterize the relationship between the positioning errors and CAs.
- We propose a novel node selection approach, and formulate the derived PEB as cost function. Avoiding the heuristic algorithms, the CAs are selected by an iterative greedy algorithm. The proposed approach can greatly decrease the number of CAs participating in the localization, at a cost of only slight accuracy degradation.
- We derive the efficient closed-form solution of AOA based CL and propose an iterative constrained Least Squares (LS) method to further refine the estimations. The proposed method considers the measurement noise at each iteration to address the problem of low iteration accuracy.

The rest of this paper is structured as follows. Section 2 gives some related work of node selection and AOA localization methods in cooperative network. Section 3 introduces the node selection approach based on PEB. Section 4 derives the closed-form solution of the AOA CL and proposes iterative constrained LS method to improve localization accuracy. Section 5 conducts extensive simulations in both 2D and 3D space to verify the effectiveness of the proposed algorithms. Finally, section 6 summarizes the main conclusions.

2. Related work

2.1. Node selection approach

Node selection approach can be achieved by blocking the broadcast of some less useful nodes and selecting the most informative nodes to reduce the energy consumption and network traffic in localization network. The CRLB matrix and the Fisher Information Matrix (FIM) are commonly used as evaluation standards with respect to the node selection scheme [4]. Specifically, two CRLB-based optimization criterions have been widely used, one for the A-optimality (minimizing the trace of CRLB matrix) and one for the D-optimality (maximizing the determinant of FIM).

In [5], the node selection strategy for RSS based CL has been proposed to catch or discard the cooperative nodes by optimizing the A-optimality criterion. Considering the measurement errors caused by Non-Line-of-Sight (NLOS), the node selection schemes by optimizing the A-optimality criterion for mixed NLOS and LOS scenario have been proposed in [6]. In [7], the authors have derived the Robust Error Bound (REB) by using CRLB, where the measurement errors have been considered. Besides, equal-REB has been used to determine the researching region based on desired localization accuracy. The aforementioned approaches assume that a single target is localized at a time and calculate the contribution of each node individually, which ignore the correlations of cooperative nodes.

In [8], the authors have formulated a cost function based on Euclidean distance and D-optimality criterion, and used local search approach to solve the node selection optimization problem. In [9], the sensor selection problem was elegantly formulated under linear measurement models by optimizing the D-optimality criterion, and solved via heuristic based on convex optimization. In [10], the authors have presented a nearly optimal sensors selection methods based on the original TDOA and transformed TOA, using A-optimality criterion under the condition of fixed number of cooperative nodes. The non-convex optimization problems with a

Boolean vector as the selection variable have been relaxed as convex Semidefinite Programming (SDP) by applying semidefinite relaxation. In [11], the authors have achieved the optimal sensor activations by optimizing the A-optimality criterion, where the correlated measurement noise is considered. The near-optimal solutions have been found by a convex relaxation approach and greedy algorithm. In [12], the authors have addressed the Joint Sensor Selection and Placement (JSSP) problem in TDOA localization, and formulated the optimization problem based on A-optimality criterion. Then, the block enumerative comparison algorithm and iterative swapping greedy algorithm have been proposed to approach the sub-optimal JSSP solutions. The aforementioned optimization algorithms are developed to search the near-optimal solution, such as SDP, genetic algorithm. Nevertheless, the approaches suffer from high computational complexity and occurrence of local optima. Thus, one purpose of this paper is to design a novel node selection approach with low computational complexity for real-time applications.

2.2. AOA localization method

Aimed to improve the localization accuracy, the AOA localization method has been studied extensively. Generally, the localization problem can be formulated as an optimization problem, and solved by LS algorithm. In [13], the authors have provided approximate expression of the variances of LS methods based TOA/AOA localization in 3D environment. In [14], the authors have derived the closed-form solution of TOA/AOA localization method based on Weighted Least Squares (WLS), whose estimator can achieve the CRLB performance in the case of small noise. Similarly, the authors in [15] have addressed the 3D localization problem by using 1D AOA measurements, and then derived the closed-form solution based on WLS estimator. In [16], the authors have proposed a two-step error variance based WLS in AOA localization network. The first step used WLS to roughly estimate the location of target. The second step used WLS again to refine the estimation, where the weight matrix was calculated from errors based on the first step. In [17], the authors have proposed the improved algorithm based on [16]. In [18], the authors have linearized the AOA measurement models and formulated a sub-optimal estimator by utilizing the convenient nature of spherical coordinates, and then used WLS to solve the problem. Although the algorithm is computationally light, the estimator cannot achieve the CRLB performance. In [19], the authors have approximated the non-convex LS estimator into a convex one by using second-order cone programming method for cooperative RSS/AOA localization. Although the several iteration-based methods were developed based on the closed-form solution [13–19], the solutions will suffer from noisy measurements.

In order to address the problem of the accuracy degradation of TDOA/AOA localization in the case of large noise as the AOA measurements are coupled in the traditional iteration equation, the authors in [20,21] proposed an iterative constrained WLS method to minimize the error. Based on it, the other purpose of this paper is to investigate an iterative constrained LS method for AOA CL method to solve the problem of low accuracy due to iterations.

3. PEB based node selection approach

3.1. Node selection problem

Consider an AOA CL network in a N -dimensional space ($N=2$ or 3), consisting of N_t agents and N_a anchors. Let $\mathcal{N}_t = \{1, 2, \dots, N_t\}$ denote the set of the agents, whose locations are to be estimated through some CL algorithms, using the AOA measurements between agents and anchors as well as the agent-to-agent measurements. Let $\mathcal{N}_a = \{N_t + 1, N_t + 2, \dots, N_t + N_a\}$ denote the set of the

anchors, whose locations are known. Since it is unrealistic to cooperate with all agents in large-scale network, node selection is necessary.

The task of node selection is to select the best subset of agents to participate in CL to minimize the estimation error. Assume that K CAs are selected from the existing network, we define a Boolean vector to represent the node selection scheme of agent k ,

$$\mathbf{s}_k = [s_k^1, s_k^2, \dots, s_k^{N_t}], \quad s_k^j \in \{0, 1\}, \quad (1)$$

where s_k^j indicates whether or not the agent j is selected as the CA. If the agent j ($j \neq k$) is selected then $s_k^j = 1$, otherwise $s_k^j = 0$. The length of the vector $\mathbf{s}_k$ is N_t and the number of the non-zero entries is K .

3.2. Signal model

Consider each node equipped with P and $P \times Q$ elements uniform linear array in 2D and 3D localization, respectively. For the sake of simplicity, define a tuple set $\mathcal{E} = \{(k, j) : k \in \mathcal{N}_t, j \in \mathcal{N}_a \cup \mathcal{N}_k^{\text{sel}}\}$ to denote the existence of an edge between two nodes, where $\mathcal{N}_k^{\text{sel}}$ denotes the set of CAs selected. The locations of all the agents are defined as $\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_t}]$. Let $\mathbf{x}_k = [x_k, y_k, z_k]$ denotes the location of agent k , and $\mathbf{p}_j = [x_j, y_j, z_j]$ denotes the location of node j (either an anchor or agent). The received waveform at time t and for agent k from the node j can be written as [22]

$$\mathbf{r}_{kj}(t) = A_{kj} \mathbf{H}_{kj} s(t - \tau_{kj}) + \mathbf{n}_{kj}(t), \quad (2)$$

where $s(t)$ denotes known waveform whose Fourier transform is denoted by $S(f)$, A_{kj} and τ_{kj} denote the complex gain and propagation delay, respectively, $\mathbf{n}_{kj}(t)$ has the complex white Gaussian variables with zero mean and two-side power spectral density $N_0/2$. For 2D localization case,

$$\mathbf{H}_{kj} = [1, e^{j\Delta \cos \varphi_{kj}}, \dots, e^{j\Delta(P-1) \cos \varphi_{kj}}]. \quad (3)$$

For 3D localization case,

$$\mathbf{H}_{kj} = \mathbf{H}_{kj}^y \odot \mathbf{H}_{kj}^x, \quad (4)$$

where φ_{kj} and α_{kj} are the azimuth and elevation angle, respectively, $\Delta = 2\pi d_a/\lambda$, d_a and λ are the antenna element separation and wavelength, respectively, $\odot$ denotes the Khatri-Rao product [23], $\mathbf{H}_{kj}^x$ and $\mathbf{H}_{kj}^y$ are the steering vectors on x -axis and y -axis, $\mathbf{H}_{kj}^x = [1, e^{i\Delta \cos \varphi_{kj} \cos \alpha_{kj}}, \dots, e^{i\Delta(P-1) \cos \varphi_{kj} \cos \alpha_{kj}}]$, and $\mathbf{H}_{kj}^y = [1, e^{i\Delta \sin \varphi_{kj} \cos \alpha_{kj}}, \dots, e^{i\Delta(Q-1) \sin \varphi_{kj} \cos \alpha_{kj}}]$.

Considering the cooperation, we introduce $\mathbf{r}$ as the vector representation of all received waveforms,

$$\mathbf{r} = [\mathbf{r}_1^T, \mathbf{r}_2^T, \dots, \mathbf{r}_{N_t}^T], \quad (5)$$

where $\mathbf{r}_k = [\mathbf{r}_k^C, \mathbf{r}_k^A]$, $\mathbf{r}_k^A$ and $\mathbf{r}_k^C$ define the received waveforms from anchors and CAs for agent k , respectively, $\mathbf{r}_k^A = [\mathbf{r}_{k(N_t+1)}^T(t), \mathbf{r}_{k(N_t+2)}^T(t), \dots, \mathbf{r}_{k(N_t+N_a)}^T(t)]$,

$\mathbf{r}_k^C = \mathbf{s}_k \circ [\mathbf{r}_{k1}^T(t), \mathbf{r}_{k2}^T(t), \dots, \mathbf{r}_{kN_t}^T(t)]$, $\circ$ is the Hadamard product.

3.3. PEB analysis

Let $\hat{\mathbf{x}}$ denote an estimate of the parameter vector $\mathbf{x}$ based on observations $\mathbf{r}$. The Mean Squared Error (MSE) matrix of $\hat{\mathbf{x}}$ satisfies the information inequality [24],

$$\mathbb{E}_{\mathbf{r}, \mathbf{x}} \{(\hat{\mathbf{x}} - \mathbf{x})(\hat{\mathbf{x}} - \mathbf{x})^T\} \geq \mathbf{J}_{\mathbf{x}}^{-1} = \text{CRLB}(\mathbf{x}), \quad (6)$$

where $\mathbf{J}_{\mathbf{x}}$ denotes the FIM for unknown parameter vector $\mathbf{x}$, whose inverse matrix is CRLB($\mathbf{x}$). The FIM is always treated as a measure

of the information about the unknown parameter vector that contained in the observations and it can be calculated by

$$\mathbf{J}_{\mathbf{x}} = \mathbb{E}_{\mathbf{r}, \mathbf{x}} \left\{ \left[\frac{\partial}{\partial \mathbf{x}} \ln f(\mathbf{r}, \mathbf{x}) \right] \left[\frac{\partial}{\partial \mathbf{x}} \ln f(\mathbf{r}, \mathbf{x}) \right]^T \right\}, \quad (7)$$

where $f(\mathbf{r}, \mathbf{x})$ is the joint Probability Density Function (PDF).

Therefore, in order to measure the localization accuracy, the following PEB is considered,

$$p(\mathbf{x}) = \sqrt{\text{tr}\{\text{CRLB}(\mathbf{x})\}} = \sqrt{\text{tr}\{\mathbf{J}_{\mathbf{x}}^{-1}\}}. \quad (8)$$

Due to the dynamic nature of the propagation channel and the inherent temporal and spatial independence of signal, we assume that the received waveforms from different nodes are independent. Based on (2), we can obtain

$$f(\mathbf{r}, \mathbf{x}) = \prod_{k \in \mathcal{N}_t} \prod_{j \in \mathcal{N}_a \cup \mathcal{N}_k^{\text{sel}}} f(\mathbf{r}_{kj} | \mathbf{x}), \quad (9)$$

where $f(\mathbf{r}_{kj} | \mathbf{x}) \propto \exp \left\{ \frac{2}{N_0} \int A_{kj} \mathbf{r}_{kj}^*(t) \mathbf{H}_{kj} s(t - \tau_{kj}) dt \right\}$. Thus, the FIM for AOA CL can be calculated by

$$\mathbf{J}_{\mathbf{x}} = \mathbf{J}_{\mathbf{x}}^A + \mathbf{J}_{\mathbf{x}}^C, \quad (10)$$

where $\mathbf{J}_{\mathbf{x}}^A$ and $\mathbf{J}_{\mathbf{x}}^C$ are the FIM from anchors and CAs, respectively.

$$\mathbf{J}_{\mathbf{x}}^A = \text{diag}(\mathbf{J}_1^A, \mathbf{J}_2^A, \dots, \mathbf{J}_k^A, \dots, \mathbf{J}_{N_t}^A), \quad (11)$$

where

$$\mathbf{J}_k^A = \sum_{j \in \mathcal{N}_a} \mathbf{G}_{kj} \mathbf{\Lambda}_{kj} \mathbf{G}_{kj}^T. \quad (12)$$

For 2D localization case,

$$\begin{cases} \mathbf{G}_{kj} = \frac{1}{d_{kj}^{2D}} [\sin^2 \varphi_{kj}, -\sin \varphi_{kj} \cos \varphi_{kj}] \\ \mathbf{\Lambda}_{kj} = \frac{1}{3} \Delta^2 (P-1) P (2P-1) \text{SNR}_{kj} \end{cases} \quad (13)$$

For 3D localization case,

$$\begin{cases} \mathbf{G}_{kj} = \frac{1}{d_{kj}^{3D}} \begin{bmatrix} 1 - \cos^2 \varphi_{kj} \cos^2 \alpha_{kj} & -\sin \varphi_{kj} \cos \varphi_{kj} \cos^2 \alpha_{kj} \\ -\sin \varphi_{kj} \cos \varphi_{kj} \cos^2 \alpha_{kj} & 1 - \sin^2 \varphi_{kj} \cos^2 \alpha_{kj} \\ -\cos \varphi_{kj} \sin \alpha_{kj} \cos \alpha_{kj} & -\sin \varphi_{kj} \sin \alpha_{kj} \cos \alpha_{kj} \end{bmatrix} \\ \mathbf{\Lambda}_{kj} = \Delta^2 \text{SNR}_{kj} Q P \begin{bmatrix} \frac{1}{3} (P-1) (2P-1) & \frac{1}{3} (P-1) (Q-1) \\ \frac{1}{3} (P-1) (Q-1) & \frac{1}{3} (Q-1) (2Q-1) \end{bmatrix} \end{cases} \quad (14)$$

where $d_{kj}^{2D} = \|(x_k, y_k) - (x_j, y_j)\|$ and $d_{kj}^{3D} = \|(x_k, y_k, z_k) - (x_j, y_j, z_j)\|$ denote the distance in 2D and 3D space, respectively, $\text{SNR}_{kj} = \int |A_{kj}|^2 |S(f)|^2 df / N_0$, P and Q are the number of the array elements on x -axis and y -axis, respectively.

Similarly, we can derive the FIM from CAs,

$$\mathbf{J}_{\mathbf{x}}^C = \begin{bmatrix} \sum_{j \in \mathcal{N}_t / \{1\}} \mathbf{J}_{1j}^C & -\mathbf{J}_{12}^C & \cdots & -\mathbf{J}_{1N_t}^C \\ -\mathbf{J}_{12}^C & \sum_{j \in \mathcal{N}_t / \{2\}} \mathbf{J}_{2j}^C & \cdots & -\mathbf{J}_{2N_t}^C \\ \vdots & \vdots & \ddots & \vdots \\ -\mathbf{J}_{1N_t}^C & -\mathbf{J}_{2N_t}^C & \cdots & \sum_{j \in \mathcal{N}_t / \{N_t\}} \mathbf{J}_{N_t j}^C \end{bmatrix}, \quad (15)$$

where $\mathbf{J}_{kj}^C = (s_k^j + s_j^k) \mathbf{G}_{kj} \mathbf{\Lambda}_{kj} \mathbf{G}_{kj}^T$ for $k \in \mathcal{N}_t$, $j \in \mathcal{N}_t / \{k\}$. The $\mathbf{J}_{\mathbf{x}}^A$ and $\mathbf{J}_{\mathbf{x}}^C$ are derived in Appendix A.

Based on (8) and (10), the PEB for cooperative AOA localization is

$$\begin{aligned} p(\mathbf{x}) &= \sqrt{\text{tr}\left\{\left(\mathbf{J}_{\mathbf{x}}^{\mathbf{A}} + \mathbf{J}_{\mathbf{x}}^{\mathbf{C}}\right)^{-1}\right\}} \\ &= \sqrt{\text{tr}\left\{\left(\mathbf{J}_{\mathbf{x}}^{\mathbf{A}}\right)^{-1}\right\} - \text{tr}\left\{\left(\mathbf{I} + \left(\mathbf{J}_{\mathbf{x}}^{\mathbf{A}}\right)^{-1}\mathbf{J}_{\mathbf{x}}^{\mathbf{C}}\right)^{-1}\left(\mathbf{J}_{\mathbf{x}}^{\mathbf{A}}\right)^{-1}\mathbf{J}_{\mathbf{x}}^{\mathbf{C}}\left(\mathbf{J}_{\mathbf{x}}^{\mathbf{A}}\right)^{-1}\right\}} \\ &= \sqrt{\text{tr}\left\{\left(\mathbf{J}_{\mathbf{x}}^{\mathbf{A}}\right)^{-1}\right\} - \Delta p(\mathbf{x})}, \end{aligned} \quad (16)$$

where $p(\mathbf{x})$ is obtained from the matrix inversion lemma. $\text{tr}(\mathbf{J}_{\mathbf{x}}^{\mathbf{A}})^{-1}$ is the squared PEB with no cooperation, $\Delta p(\mathbf{x})$ can be interpreted as the decrease of PEB (DPEB) due to the cooperation from agents. It can be shown that the DPEB depends on $\mathbf{J}_{\mathbf{x}}^{\mathbf{C}}$ and $\mathbf{J}_{\mathbf{x}}^{\mathbf{A}}$. $\mathbf{J}_{\mathbf{x}}^{\mathbf{C}}$ is highly interrelated among agents, while $\mathbf{J}_{\mathbf{x}}^{\mathbf{A}}$ is not interrelated. This is intuitive since the effectiveness of the localization information provided by the CAs depends on their position errors (i.e., position uncertainties).

3.4. Implementation of node selection

Based (15) and (16), we can formulate the node selection problem as the following optimization problem:

$$\begin{aligned} &\text{maximize } \Delta p(\mathbf{x}) \\ &\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_{N_t}\} \end{aligned} \quad (17)$$

$$\text{s.t. C1: } \mathbf{1}^T \mathbf{s}_k = K, \quad (18)$$

$$\text{C2: } \mathbf{s}_k \in \{0, 1\}^{N_t}, \quad (19)$$

$$\text{C3: } s_k^j = 0 \text{ for } k = j. \quad (20)$$

In this problem, $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_{N_t}\}$ are the Boolean vectors, which can be regarded as node selection scheme for all the agents. The objective function is the DPEB due to cooperation determined by s_k^j . The constraints C1-C3 denote that each Boolean vector tries to select K CAs from $(N_t - 1)$ agents. Since the optimization problem is an integer programming problem with a Boolean vector as the optimization variable, it is NP-hard. The simplest and most accurate method is the exhaustive search method, which is, however, not suitable for large network due to its prohibitively expensive complexity.

In order to solve the problem, we proposed an iterative greedy node selection approach based on PEB, which can obtain suboptimal solutions with low computational complexity. Firstly, we investigate the DPEB if a new CA j enters the localization network. If $\mathbf{s}_k^{(l-1)} = [s_k^1, s_k^2, \dots, s_k^{N_t}]$ and $\mathbf{s}_k^{(l)} = [\tilde{s}_k^1, \tilde{s}_k^2, \dots, \tilde{s}_k^{N_t}]$ represent the node selection schemes of agent k at iteration $(l-1)$ and l , respectively, where $\tilde{s}_k^i = s_k^i$ for $i \in \mathcal{N}_t \setminus \{j\}$, $s_k^j = 0$ and $\tilde{s}_k^j = 1$, then the resulting FIM satisfies $\mathbf{J}_k^l \geq \mathbf{J}_k^{(l-1)}$. More precisely,

$$\mathbf{J}_k^l - \mathbf{J}_k^{(l-1)} = \mathbf{G}_{kj} \mathbf{A}_{kj}^* \mathbf{G}_{kj}^T, \quad (21)$$

and

$$\begin{aligned} \Delta p_{kj} &= \text{tr}\left\{\left(\mathbf{J}_k^{(l-1)}\right)^{-1}\right\} - \text{tr}\left\{\left(\mathbf{J}_k^l\right)^{-1}\right\}, \\ &= \text{tr}\left\{\left(\mathbf{J}_k^{(l-1)}\right)^{-1} \mathbf{G}_{kj} \left[\left(\mathbf{A}_{kj}^*\right)^{-1} + \mathbf{G}_{kj}^T \left(\mathbf{J}_k^{(l-1)}\right)^{-1} \mathbf{G}_{kj}\right]^{-1} \mathbf{G}_{kj}^T \left(\mathbf{J}_k^{(l-1)}\right)^{-1}\right\}. \end{aligned} \quad (22)$$

where

$$\mathbf{A}_{kj}^* = \mathbf{A}_{kj} \left(\mathbf{I} - \mathbf{Z}_{kj} \mathbf{A}_{kj} + \mathbf{Z}_{kj} \left(\mathbf{A}_{kj}^{-1} + \mathbf{Z}_{kj} \right)^{-1} \mathbf{Z}_{kj}^T \mathbf{A}_{kj} \right), \quad (23)$$

$$\text{and } \mathbf{Z}_{kj} = \mathbf{G}_{kj}^T \left(\mathbf{J}_j^{\mathbf{A}} \right)^{-1} \mathbf{G}_{kj}.$$

It is clear that when a new CA enters localization network, the increase in Fisher information leads to the DPEB (Δp_{kj}). Δp_{kj} is related to geometric configuration ($\mathbf{G}_{kj}$) and position uncertainties of CAs ($\mathbf{J}_j^{\mathbf{A}}$). When the position of the CA j is precisely known ($\mathbf{J}_j^{\mathbf{A}} \rightarrow \infty$), Δp_{kj} will take its maximum value, $\Delta p_{kj} = \text{tr}\left\{\left(\mathbf{J}_k^{(l-1)}\right)^{-1} \mathbf{G}_{kj} \left[\left(\mathbf{A}_{kj}\right)^{-1} + \mathbf{G}_{kj}^T \left(\mathbf{J}_k^{(l-1)}\right)^{-1} \mathbf{G}_{kj}\right]^{-1} \mathbf{G}_{kj}^T \left(\mathbf{J}_k^{(l-1)}\right)^{-1}\right\}$.

When the position of the CA j is unavailable ($\mathbf{J}_j^{\mathbf{A}} \rightarrow 0$), Δp_{kj} will take its minimum value, $\Delta p_{kj} = 0$.

In the proposed approach, we iteratively select a new CA which provides the largest performance improvement until the number of CAs is equal to K . Clearly, at each iteration of the greedy algorithm, the newly entering CA is the one that maximizes the performance improvement characterized by Δp_{kj} . The iterative greedy node selection approach is summarized in Algorithm 1.

Algorithm 1 iterative greedy node selection approach based on PEB.

Require: anchors' locations, array parameters: P, Q, d_a, λ .

Ensure: $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_{N_t}$.

- 1: Calculate $\mathbf{J}_{\mathbf{x}}^{\mathbf{A}} = \text{diag}(\mathbf{J}_1^{\mathbf{A}}, \dots, \mathbf{J}_k^{\mathbf{A}}, \dots, \mathbf{J}_{N_t}^{\mathbf{A}})$ by Eq. (12);
- 2: **for** $k \in \mathcal{N}_t$ **do**
- 3: initialize $\mathbf{s}_k = [0, 0, \dots, 0]$, $\mathbf{J}_k^0 = \mathbf{J}_k^{\mathbf{A}}$, $\mathcal{L} = \mathcal{N}_t \setminus \{k\}$;
- 4: **for** $l = 1 : K$ **do**
- 5: enumerate all agents in $\mathcal{L}$ to determine node $j \in \mathcal{L}$ such that Δp_{kj} is maximized;
- 6: update $\mathbf{s}_k$ by setting $s_k^j = 1$, and update $\mathbf{J}_k^l$ by adding $\mathbf{G}_{kj} \mathbf{A}_{kj}^* \mathbf{G}_{kj}^T$;
- 7: remove node j from $\mathcal{L}$;
- 8: **end for**
- 9: **end for**

4. CL with LS estimation

Based on the node selection scheme, this section aims to simultaneously determine the locations of all agents with respect to the observations. We derive an efficient closed-form solution of AOA based CL and propose an iterative constrained LS method to refine the estimations.

4.1. Closed-form solution

In AOA localization network, the antenna arrays equipped at nodes usually measure a function of AOA, instead of AOA itself. Therefore, the AOA measurement in 2D space can be modeled as $\hat{\omega}(\varphi_{kj})$ [25].

$$\hat{\omega}(\varphi_{kj}) = \cos \varphi_{kj} + n_{kj}, \text{ for } (k, j) \in \mathcal{E}, \quad (24)$$

where n_{kj} is the measurement error of $\cos \varphi_{kj}$, modeled as a zero-mean Gaussian random variable with $\sigma_{n_{kj}}^2$.

$$\hat{\varphi}_{kj} = \tan^{-1} \left(\frac{y_k - y_j}{x_k - x_j} \right) + \kappa_{kj}, \quad (25)$$

where $\hat{\varphi}_{kj}$ denotes the estimation of azimuth angle, and κ_{kj} is the bias. When the bias is small, $\cos \kappa_{kj} \approx 1$, $\sin \kappa_{kj} \approx \kappa_{kj}$, and $\kappa_{kj} = \frac{n_{kj}}{\sin \varphi_{kj}}$.

According to Tomic et al. [18], (25) is rewritten as $\mathbf{a}_{kj}^T(\mathbf{x}_k - \mathbf{p}_j) = 0$, where $\mathbf{a}_{kj} = [-\sin \varphi_{kj}, \cos \varphi_{kj}]$. Applying the LS approach to (25), result in

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \sum_{(k,j) \in \mathcal{E}} (\mathbf{a}_{kj}^T(\mathbf{x}_k - \mathbf{p}_j))^2. \quad (26)$$

Notice that all the sums are quadratic with respect to $\mathbf{x}$. Hence, (26) can be rewritten in an equivalent vector form as

$$\text{minimize } \|\mathbf{Ax} - \mathbf{b}\|^2, \quad (27)$$

where $\mathbf{A} \in \mathbb{R}^{|\mathcal{E}|N_a \times 2N_t}$,

$$\begin{aligned} \mathbf{A}_{q,(2k-1):2k} &= \mathbf{a}_{kj}^T, \mathbf{b}_q = \mathbf{a}_{kj}^T \mathbf{p}_j \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|, \\ \mathbf{A}_{q,[(2k-1):2k, (2j-1):2j]} &= [\mathbf{a}_{kj}^T, -\mathbf{a}_{kj}^T] \mathbf{b}_q = 0 \text{ for } (k,j) \in \mathcal{E}_C, q = |\mathcal{E}_A| + 1, \dots, |\mathcal{E}|, \end{aligned}$$

$\mathcal{E}_A$ and $\mathcal{E}_C$ denote the tuple sets of all agent/anchor and agent/CA edges, respectively, $|\bullet|$ denotes the cardinality of the set. Then, the solution of (27) is

$$\hat{\mathbf{x}} = \mathbf{A}^\dagger \mathbf{b}, \quad (28)$$

where $\dagger$ stands for the Moore-Penrose pseudo-inverse of matrix.

In order to refine localization results, an iteration inversion algorithm is employed, and we have the following Taylor series expansion,

$$\mathbf{f}(\mathbf{x}) - \mathbf{f}(\hat{\mathbf{x}}) = \mathbf{R} \Delta \mathbf{x}, \quad (29)$$

$$\text{where } \mathbf{f}(\mathbf{x}) = [\mathbf{f}_1(\mathbf{x}), \mathbf{f}_2(\mathbf{x}), \dots, \mathbf{f}_{N_t}(\mathbf{x})], \quad \mathbf{f}_k(\mathbf{x}) = [f_{k1}(\mathbf{x}), \dots, f_{k|\mathcal{E}|}(\mathbf{x})],$$

$$f_{kj}(\mathbf{x}) = \hat{w}(\varphi_{kj}) - \cos \varphi_{kj} \text{ for } (k,j) \in \mathcal{E}_A, \text{ and } f_{kj}(\mathbf{x}) = 0 \text{ for } (k,j) \in \mathcal{E}_C. \mathbf{R} \text{ is the Jacobian matrix, i.e.,}$$

$$\mathbf{R} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}^T} \bigg|_{\mathbf{x}=\hat{\mathbf{x}}}, \quad (30)$$

$$\text{where } \mathbf{R} \in \mathbb{R}^{|\mathcal{E}|N_t \times 2N_t}, \quad \mathbf{R}_{q,(2k-1):2k} = \mathbf{h}_{kj} \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|,$$

$$\mathbf{R}_{q,[(2k-1):2k, (2j-1):2j]} = [\mathbf{h}_{kj}, -\mathbf{h}_{kj}] \text{ for } (k,j) \in \mathcal{E}_C, q = |\mathcal{E}_A| + 1, \dots, |\mathcal{E}|, \text{ and } \mathbf{h}_{kj} = \left[\frac{\sin^2 \varphi_{kj}}{d_{kj}}, -\frac{\sin \varphi_{kj} \cos \varphi_{kj}}{d_{kj}} \right]^T.$$

Since measurement errors are always very small, $\mathbf{f}(\mathbf{x}) \approx \mathbf{0}_{|\mathcal{E}|N_t \times 1}$. Thus, $\Delta \mathbf{x}$ can be obtained by

$$\Delta \mathbf{x} = -(\mathbf{R}^T \mathbf{R})^{-1} \mathbf{R}^T \mathbf{f}(\hat{\mathbf{x}}). \quad (31)$$

Thus, the location estimation of the agents can be refined as

$$\hat{\mathbf{x}}_r = \hat{\mathbf{x}} + \Delta \mathbf{x}. \quad (32)$$

4.2. Improved iterative constrained LS method

Based on (32), iterative LS can be used to improve localization accuracy. However, since the noisy measurements are coupled in $\mathbf{R}$, the estimator of the closed form solution will suffer from performance degradation at high noise levels. In order to further improve the localization accuracy, we will take into account the measurement noise in $\mathbf{R}$.

We first introduce the augmented matrix $\mathbf{B} = [\mathbf{R}, \mathbf{f}(\hat{\mathbf{x}})]^T$ and $\mathbf{V} = [\Delta \mathbf{x}, 1]$. Then, we define the cost function with the augmented matrix as

$$J(\Delta \mathbf{x}) = \text{minimize } (\mathbf{BV})^T \mathbf{BV}. \quad (33)$$

Considering the measurement noise, $\mathbf{B}$ can be composed as

$$\mathbf{B} = \mathbf{B}^0 + \Delta \mathbf{B}, \quad (34)$$

where $\mathbf{B}^0$ denotes the true augmented matrix, $\Delta \mathbf{B} = [\Delta \mathbf{R}, \Delta \mathbf{f}(\hat{\mathbf{x}})]^T$ is the noise matrix, which can be reformulated as

$$\Delta \mathbf{B} = \begin{bmatrix} \mathbf{C}_1 & \mathbf{C}_2 \end{bmatrix}, \quad (35)$$

where $\mathbf{C}_1 \in \mathbb{R}^{|\mathcal{E}|N_t \times 2N_t}$, $\mathbf{C}_2 \in \mathbb{R}^{|\mathcal{E}|N_t \times 1}$, let $\mathbf{c}_{kj} = \frac{1}{d_{kj}} [\sin 2\varphi_{kj}, -\cos 2\varphi_{kj}]$,

$$[\mathbf{C}_1]_{q,(2k-1):2k} = \kappa_{kj} \mathbf{c}_{kj}^T \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|,$$

$$[\mathbf{C}_1]_{q,[(2k-1):2k, (2j-1):2j]} = \kappa_{kj} [\mathbf{c}_{kj}^T, -\mathbf{c}_{kj}^T] \text{ for } (k,j) \in \mathcal{E}_C, q = |\mathcal{E}_A| + 1, \dots, |\mathcal{E}|,$$

$$[\mathbf{C}_2]_{q,1} = \kappa_{kj} \sin \varphi_{kj} \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|,$$

$$[\mathbf{C}_2]_{q,1} = 0 \text{ for } (k,j) \in \mathcal{E}_C, q = |\mathcal{E}_A| + 1, |\mathcal{E}_A| + 2, \dots, |\mathcal{E}|.$$

Substituting (34) into (33) yields the following cost function:

$$J(\Delta \mathbf{x}) = \mathbf{V}^T \mathbf{B}^0 T \mathbf{B}^0 \mathbf{V} + \mathbf{V}^T \Delta \mathbf{B}^T \Delta \mathbf{B} \mathbf{V} + 2\mathbf{V}^T \Delta \mathbf{B}^T \mathbf{B}^0 \mathbf{V}. \quad (36)$$

We take expectation of J and the third term on the right side of (36) is zero-mean. Therefore,

$$\mathbb{E}\{J(\Delta \mathbf{x})\} = \mathbf{V}^T \mathbf{B}^0 T \mathbf{B}^0 \mathbf{V} + \mathbf{V}^T \mathbb{E}\{\Delta \mathbf{B}^T \Delta \mathbf{B}\} \mathbf{V}. \quad (37)$$

Based on [26], we treat the second term on the right side of (37) as a constant constraint. Then, the cost function can be rewritten as

$$\begin{aligned} \text{minimize } & \mathbf{V}^T \mathbf{B}^T \mathbf{B} \mathbf{V}, \\ \text{s.t. } & \mathbf{V}^T \mathbf{\Omega} \mathbf{V} = \kappa, \end{aligned} \quad (38)$$

where $\mathbf{\Omega} = \mathbb{E}(\Delta \mathbf{B}^T \Delta \mathbf{B})$, the constant κ can be any positive value and its value causes only a different scaling of $\mathbf{V}$.

Next, we use Lagrange multiplier to solve the constrained minimization problem. (38) can be rewritten as

$$\text{minimize } \mathbf{V}^T \mathbf{B}^T \mathbf{B} \mathbf{V} - \lambda (\kappa - \mathbf{V}^T \mathbf{\Omega} \mathbf{V}). \quad (39)$$

By taking its derivative with respect to $\mathbf{V}$ and setting it to zero,

$$\mathbf{B}^T \mathbf{B} \mathbf{V} = \lambda \mathbf{\Omega} \mathbf{V}. \quad (40)$$

Based on the generalized singular value decomposition, the solution vector $\mathbf{V}$ is the generalized eigenvector, which corresponds to minimum generalized eigenvalue of the pair $(\mathbf{B}^T \mathbf{B} \text{ and } \mathbf{\Omega})$. Therefore, the estimation error $\Delta \mathbf{x}$ is

$$\Delta \mathbf{x} = \frac{\mathbf{V}(1 : 2N_t)}{\mathbf{V}(2N_t + 1)}. \quad (41)$$

The complete description of AOA CL based on iterative constrained LS is given in Algorithm 2.

Algorithm 2 AOA CL based on iterative constrained LS.

Require: anchors' locations, AOA measurements, $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_{N_t}$;

Ensure: $\hat{\mathbf{x}}_u$

- 1: Coarse estimations by solving LS problem (28) $\hat{\mathbf{x}}$ can be obtained;
 - 2: Localization results can be refined by (31), and the closed-form solution $\hat{\mathbf{x}}_r$ can be obtained in (32);
 - 3: Define a convergence threshold ζ_c and a maximum number of iterations n_{iter} ;
 - 4: Set $l = 0$ and $\hat{\mathbf{x}}^0 = \hat{\mathbf{x}}_r$;
 - 5: **repeat**
 - 6: Solve problem (33) to obtain $\Delta \mathbf{x}$;
 - 7: $\hat{\mathbf{x}}^{l+1} = \hat{\mathbf{x}}^l + \Delta \mathbf{x}$;
 - 8: $\zeta = \|\hat{\mathbf{x}}^{l+1} - \hat{\mathbf{x}}^l\|$;
 - 9: $l = l + 1$;
 - 10: **until** $\zeta < \zeta_c$ or $l > n_{\text{iter}}$;
 - 11: $\hat{\mathbf{x}}_u = \hat{\mathbf{x}}^{l+1}$;
-

4.3. Extension to 3D space

Similarly, the measurements in 3D space can be modeled as $\hat{w}(\varphi_{kj}, \alpha_{kj})$ and $\hat{v}(\varphi_{kj}, \alpha_{kj})$, respectively.

$$\begin{cases} \hat{w}(\varphi_{kj}, \alpha_{kj}) = \cos \varphi_{kj} \cos \alpha_{kj} + \mu_{kj}, & \text{for } (k,j) \in \mathcal{E}, \\ \hat{v}(\varphi_{kj}, \alpha_{kj}) = \sin \varphi_{kj} \cos \alpha_{kj} + \nu_{kj} \end{cases} \quad (42)$$

where μ_{kj} is the measurement error of $\cos \varphi_{kj} \cos \alpha_{kj}$, modeled as a zero-mean Gaussian random variable with $\sigma_{\mu_{kj}}^2$, ν_{kj} is the measurement error of $\sin \varphi_{kj} \cos \alpha_{kj}$, modeled as a zero-mean Gaussian random variable with $\sigma_{\nu_{kj}}^2$.

$$\hat{\varphi}_{kj} = \tan^{-1} \left(\frac{y_k - y_j}{x_k - x_j} \right) + \kappa_{\varphi_{kj}}, \hat{\alpha}_{kj} = \sin^{-1} \left(\frac{z_k - z_j}{\|\mathbf{x}_k - \mathbf{p}_j\|} \right) + \kappa_{\alpha_{kj}}, \quad (43)$$

where $\kappa_{\varphi_{kj}}$ and $\kappa_{\alpha_{kj}}$ denote the biases of φ_{kj} and α_{kj} , respectively, when the biases are small, $\kappa_{\varphi_{kj}} = \frac{1}{\cos \alpha_{kj}} (-\mu_{kj} \sin \varphi_{kj} + \nu_{kj} \cos \varphi_{kj})$, and $\kappa_{\alpha_{kj}} = \frac{1}{\sin \alpha_{kj}} (\mu_{kj} \cos \varphi_{kj} + \nu_{kj} \sin \varphi_{kj})$. Applying the 3D parameters to (43),

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \sum_{(k,j) \in \mathcal{E}} (\mathbf{v}_{kj}^T (\mathbf{x}_k - \mathbf{p}_j))^2 + \sum_{(k,j) \in \mathcal{E}} \left((\sin \alpha_{kj} \mathbf{u}_{kj} - \mathbf{k})^T (\mathbf{x}_k - \mathbf{p}_j) \right)^2, \quad (44)$$

where $\mathbf{v}_{kj} = [-\sin \varphi_{kj}, \cos \varphi_{kj}, 0]$, $\mathbf{u}_{kj} = [\cos \varphi_{kj} \cos \alpha_{kj}, \sin \varphi_{kj} \cos \alpha_{kj}, \sin \alpha_{kj}]$, $\mathbf{k} = [0, 0, 1]$. Thus, (27) can be updated as

$$\min_{\mathbf{x}} \|\mathbf{A}_{3D} \mathbf{x} - \mathbf{b}_{3D}\|^2, \quad (45)$$

where $\mathbf{A}_{3D} \in \mathbb{R}^{2|\mathcal{E}|N_t \times 3N_t}$,

$$[\mathbf{A}_{3D}]_{q,(3k-2):3k} = \mathbf{v}_{kj}^T, [\mathbf{b}_{3D}]_q = \mathbf{v}_{kj}^T \mathbf{p}_j \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|,$$

$$[\mathbf{A}_{3D}]_{q,(3k-2):3k,(3j-2):3j} = (\sin \alpha_{kj} \mathbf{u}_{kj} - \mathbf{k})^T, [\mathbf{b}_{3D}]_q = (\sin \alpha_{kj} \mathbf{u}_{kj} - \mathbf{k})^T \mathbf{p}_j, \text{ for } (k,j) \in \mathcal{E}_A, q = |\mathcal{E}_A| + 1, |\mathcal{E}_A| + 2, \dots, 2|\mathcal{E}_A|,$$

$$[\mathbf{A}_{3D}]_{q,(3k-2):3k,(3j-2):3j} = [\mathbf{v}_{kj}^T, -\mathbf{v}_{kj}^T], [\mathbf{b}_{3D}]_q = 0, \text{ for } (k,j) \in \mathcal{E}_C, q = 2|\mathcal{E}_A| + 1, 2|\mathcal{E}_A| + 2, \dots, 2|\mathcal{E}_A| + |\mathcal{E}_C|,$$

$$[\mathbf{A}_{3D}]_{q,(3k-2):3k,(3j-2):3j} = [(\sin \alpha_{kj} \mathbf{u}_{kj} - \mathbf{k})^T, -(\sin \alpha_{kj} \mathbf{u}_{kj} - \mathbf{k})^T],$$

$$[\mathbf{b}_{3D}]_q = 0,$$

$$\text{for } (k,j) \in \mathcal{E}_C, q = 2|\mathcal{E}_A| + |\mathcal{E}_C| + 1, \dots, 2|\mathcal{E}|.$$

Thus, the solution of (45) is $\hat{\mathbf{x}} = \mathbf{A}_{3D}^\dagger \mathbf{b}_{3D}$. In order to refine the localization results, $\mathbf{R}$ matrix in (31) can be replaced by

$$\mathbf{R}_{3D} = \frac{\partial \mathbf{f}_{3D}}{\partial \mathbf{x}^T} \bigg|_{\mathbf{x}=\hat{\mathbf{x}}}, \quad (46)$$

where $\mathbf{f}_{3D}(\mathbf{x}) = [\mathbf{f}_1(\mathbf{x}), \mathbf{f}_2(\mathbf{x}), \dots, \mathbf{f}_{N_t}(\mathbf{x})]$, $\mathbf{f}_k(\mathbf{x}) = [f_{k1}(\mathbf{x}), \dots, f_{k(2|\mathcal{E}|)}(\mathbf{x})]$.

$$[\mathbf{f}_k(\mathbf{x})]_{q,1} = \hat{\omega}(\varphi_{kj}, \alpha_{kj}) - \cos \varphi_{kj} \cos \alpha_{kj} \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|,$$

$$[\mathbf{f}_k(\mathbf{x})]_{q,1} = \hat{\gamma}(\varphi_{kj}, \alpha_{kj}) - \sin \varphi_{kj} \cos \alpha_{kj} \text{ for } (k,j) \in \mathcal{E}_A, q = |\mathcal{E}_A| + 1, \dots, 2|\mathcal{E}_A|,$$

$$[\mathbf{f}_k(\mathbf{x})]_{q,1} = 0 \text{ for } (k,j) \in \mathcal{E}_C, q = 2|\mathcal{E}_A| + 1, \dots, 2|\mathcal{E}|.$$

$$[\mathbf{R}_{3D}]_{q,(3k-2):3k} = \mathbf{g}_{kj} \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|,$$

$$[\mathbf{R}_{3D}]_{q,(3k-2):3k} = \mathbf{y}_{kj} \text{ for } (k,j) \in \mathcal{E}_A, q = |\mathcal{E}_A| + 1, |\mathcal{E}_A| + 2, \dots, 2|\mathcal{E}_A|,$$

$$[\mathbf{R}_{3D}]_{q,(3k-2):3k,(3j-2):3j} = [\mathbf{g}_{kj}, -\mathbf{g}_{kj}] \text{ for } (k,j) \in \mathcal{E}_C, q = 2|\mathcal{E}_A| + 1, \dots, 2|\mathcal{E}_A| + |\mathcal{E}_C|,$$

$$[\mathbf{R}_{3D}]_{q,(3k-2):3k,(3j-2):3j} = [\mathbf{y}_{kj}, -\mathbf{y}_{kj}] \text{ for } (k,j) \in \mathcal{E}_C, q = 2|\mathcal{E}_A| + |\mathcal{E}_C| + 1, \dots, 2|\mathcal{E}|,$$

$$\mathbf{g}_{kj} = \begin{bmatrix} \frac{1 - \cos^2 \varphi_{kj} \cos^2 \alpha_{kj}}{d_{kj}}, & \frac{-\sin \varphi_{kj} \cos \varphi_{kj} \cos^2 \alpha_{kj}}{d_{kj}}, & \frac{-\cos \varphi_{kj} \cos \alpha_{kj} \sin \alpha_{kj}}{d_{kj}} \end{bmatrix}^T,$$

$$\mathbf{y}_{kj} = \begin{bmatrix} \frac{-\sin \varphi_{kj} \cos \varphi_{kj} \cos^2 \alpha_{kj}}{d_{kj}}, & \frac{1 - \sin^2 \varphi_{kj} \cos^2 \alpha_{kj}}{d_{kj}}, & \frac{-\sin \varphi_{kj} \cos \alpha_{kj} \sin \alpha_{kj}}{d_{kj}} \end{bmatrix}^T.$$

Then, we can obtain $\Delta \mathbf{x}$ by (31). Similarly, we define the cost function with the augmented matrix $\mathbf{B}_{3D} = [\mathbf{R}_{3D} \quad \mathbf{f}_{3D}(\hat{\mathbf{x}})]$,

$$J_{3D}(\Delta \mathbf{x}) = \minimize (\mathbf{B}_{3D} \mathbf{V}_{3D})^T \mathbf{B}_{3D} \mathbf{V}_{3D}, \quad (47)$$

where $\mathbf{B}_{3D} = \mathbf{B}_{3D}^0 + \Delta \mathbf{B}_{3D}$, $\mathbf{B}_{3D}^0$ and $\Delta \mathbf{B}_{3D}$ are the true augmented matrix and noise matrix, respectively,

$$\Delta \mathbf{B}_{3D} = \begin{bmatrix} \mathbf{C}_1^{3D} & \mathbf{C}_2^{3D} \end{bmatrix}, \quad (48)$$

where $\mathbf{C}_1^{3D} \in \mathbb{R}^{2|\mathcal{E}|N_t \times 3N_t}$, $\mathbf{C}_2^{3D} \in \mathbb{R}^{2|\mathcal{E}|N_t \times 1}$. Let $\delta_{kj} = [\kappa_{\varphi_{kj}}, \kappa_{\alpha_{kj}}]^T$, thus,

$$[\mathbf{C}_1^{3D}]_{q,(3k-2):3k} = (\mathbf{d}_{kj} \delta_{kj})^T \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|,$$

$$[\mathbf{C}_1^{3D}]_{q,(3k-2):3k} = (\mathbf{s}_{kj} \delta_{kj})^T \text{ for } (k,j) \in \mathcal{E}_A, q = |\mathcal{E}_A| + 1, \dots, 2|\mathcal{E}_A|,$$

$$[\mathbf{C}_1^{3D}]_{q,[(3k-2):3k,(3j-2):3j]} = \delta_{kj}^T [\mathbf{d}_{kj}^T, -\mathbf{d}_{kj}^T] \text{ for } (k,j) \in \mathcal{E}_C, q = 2|\mathcal{E}_A| + 1, \dots, 2|\mathcal{E}_A| + |\mathcal{E}_C|,$$

$$[\mathbf{C}_1^{3D}]_{q,[(3k-2):3k,(3j-2):3j]} = \delta_{kj}^T [\mathbf{s}_{kj}^T, -\mathbf{s}_{kj}^T] \text{ for } (k,j) \in \mathcal{E}_C, q = 2|\mathcal{E}_A| + |\mathcal{E}_C| + 1, \dots, 2|\mathcal{E}|,$$

$$\mathbf{d}_{kj} = \begin{bmatrix} \frac{\sin 2\varphi_{kj} \cos^2 \alpha_{kj}}{d_{kj}} & \frac{\cos^2 \varphi_{kj} \sin 2\alpha_{kj}}{d_{kj}} \\ \frac{-\cos 2\varphi_{kj} \cos^2 \alpha_{kj}}{d_{kj}} & \frac{\sin 2\varphi_{kj} \sin 2\alpha_{kj}}{2d_{kj}} \\ \frac{\sin \varphi_{kj} \sin 2\alpha_{kj}}{2d_{kj}} & \frac{-\cos \varphi_{kj} \cos 2\alpha_{kj}}{d_{kj}} \end{bmatrix}, \quad \mathbf{s}_{kj} = \begin{bmatrix} \frac{-\cos 2\varphi_{kj} \cos^2 \alpha_{kj}}{d_{kj}} & \frac{\sin 2\varphi_{kj} \sin 2\alpha_{kj}}{2d_{kj}} \\ \frac{-\sin 2\varphi_{kj} \cos^2 \alpha_{kj}}{d_{kj}} & \frac{\sin^2 \varphi_{kj} \sin 2\alpha_{kj}}{2d_{kj}} \\ \frac{-\cos \varphi_{kj} \sin 2\alpha_{kj}}{2d_{kj}} & \frac{-\sin \varphi_{kj} \cos 2\alpha_{kj}}{d_{kj}} \end{bmatrix},$$

$$[\mathbf{C}_2^{3D}]_{q,1} = \sin \varphi_{kj} \cos \alpha_{kj} \kappa_{\varphi_{kj}} + \cos \varphi_{kj} \sin \alpha_{kj} \kappa_{\alpha_{kj}} \text{ for } (k,j) \in \mathcal{E}_A, q = 1, 2, \dots, |\mathcal{E}_A|,$$

$$[\mathbf{C}_2^{3D}]_{q,1} = -\cos \varphi_{kj} \cos \alpha_{kj} \kappa_{\varphi_{kj}} + \sin \varphi_{kj} \sin \alpha_{kj} \kappa_{\alpha_{kj}} \text{ for } (k,j) \in \mathcal{E}_A, q = |\mathcal{E}_A| + 1, \dots, 2|\mathcal{E}_A|,$$

$$[\mathbf{C}_2^{3D}]_{q,1} = 0 \text{ for } (k,j) \in \mathcal{E}_C, q = 2|\mathcal{E}_A| + 1, \dots, 2|\mathcal{E}|.$$

Next, the Lagrange multiplier is used to solve the constrained minimization problem, and the estimation error $\Delta \mathbf{x}$ can be obtained.

4.4. Performance analysis

In (31), the noise in measurements will result in estimation error, and similar to the work in [27], the estimated covariance matrix can be approximated by

$$\text{cov}(\Delta \mathbf{x}) = (\mathbf{R}^T \mathbf{Q}^{-1} \mathbf{R})^{-1}, \quad (49)$$

where $\mathbf{Q} = \mathbb{E}(\boldsymbol{\eta} \boldsymbol{\eta}^T)$ is the covariance matrix of measurements, $\boldsymbol{\eta} = [\boldsymbol{\eta}^A, \boldsymbol{\eta}^C]$,

$$\boldsymbol{\eta}^A = [\boldsymbol{\eta}_1^A, \boldsymbol{\eta}_2^A, \dots, \boldsymbol{\eta}_{N_t}^A], \boldsymbol{\eta}_k^A = [n_{k1}, n_{k2}, \dots, n_{kN_a}].$$

$$\boldsymbol{\eta}^C = [\boldsymbol{\eta}_1^C, \boldsymbol{\eta}_2^C, \dots, \boldsymbol{\eta}_{N_t}^C], \boldsymbol{\eta}_k^C = [n_{k1}, \dots, n_{k(k-1)}, n_{k(k+1)}, \dots, n_{kN_t}].$$

Due to the independence of measurement noise, $\mathbf{Q} = \text{diag}(\mathbf{Q}_A, \mathbf{Q}_C)$. According to (30), the matrix $\mathbf{R}$ can be decomposed as

$$\mathbf{R} = [\mathbf{R}^A, \mathbf{R}^C], \quad (50)$$

where $[\mathbf{R}^A]_{q,(2k-1):2k} = \mathbf{h}_{kj}$ for $k \in \mathcal{N}_t, j \in \mathcal{N}_a, q = 1, 2, \dots, N_a$,

$$[\mathbf{R}^C]_{q,[(2k-1):2k,(2j-1):2j]} = \mathbf{s}_k^j [\mathbf{h}_{kj}, -\mathbf{h}_{kj}] \text{ for } k \in \mathcal{N}_t, j \in \mathcal{N}_t / \{k\}, q = 1, \dots, N_t - 1.$$

Thus, (49) can be rewritten as

$$\text{cov}(\Delta \mathbf{x}) = \left((\mathbf{R}^A)^T \mathbf{Q}_A^{-1} \mathbf{R}^A + (\mathbf{R}^C)^T \mathbf{Q}_C^{-1} \mathbf{R}^C \right)^{-1}, \quad (51)$$

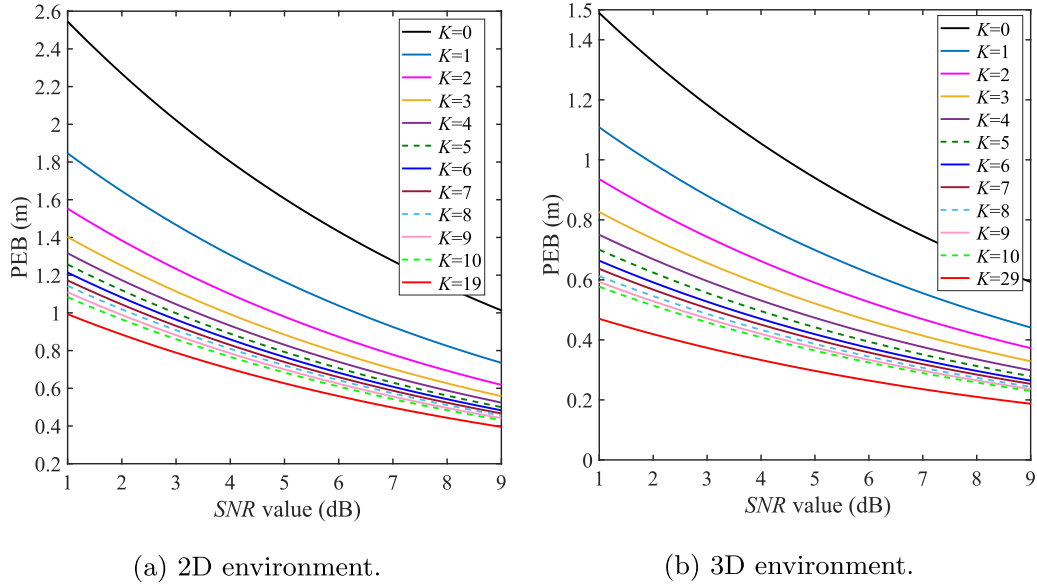


Fig. 1. PEBs for different number of CAs.

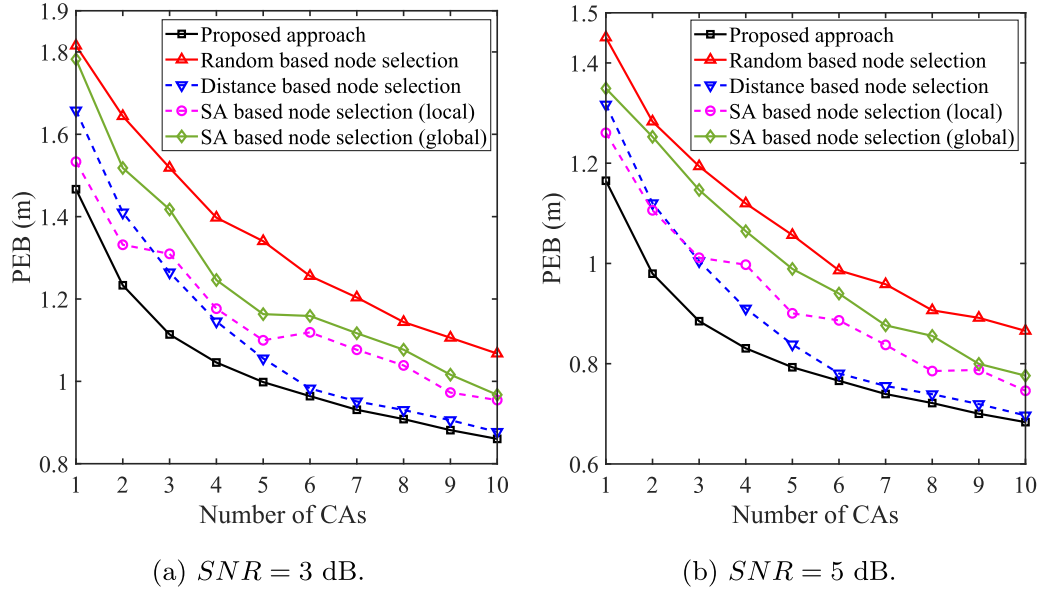


Fig. 2. PEBs for different node selection approaches in 2D environment.

where

$$\mathbf{F}_k^A = \sum_{j \in \mathcal{N}_a} (\sigma_{n_{kj}}^2)^{-1} \mathbf{h}_{kj}^T \mathbf{h}_{kj}, \quad (\mathbf{R}^A)^T \mathbf{Q}_A^{-1} \mathbf{R}^A = \text{diag}(\mathbf{F}_1^A, \dots, \mathbf{F}_k^A, \dots, \mathbf{F}_{N_t}^A),$$

$$(\mathbf{R}^C)^T \mathbf{Q}_C^{-1} \mathbf{R}^C =$$

$$\begin{bmatrix} \sum_{j \in \mathcal{N}_t / \{1\}} \mathbf{F}_{1j}^C & -\mathbf{F}_{12}^C & \cdots & -\mathbf{F}_{1N_t}^C \\ -\mathbf{F}_{12}^C & \sum_{j \in \mathcal{N}_t / \{2\}} \mathbf{F}_{2j}^C & \cdots & -\mathbf{F}_{2N_t}^C \\ \vdots & \vdots & \ddots & \vdots \\ -\mathbf{F}_{1N_t}^C & -\mathbf{F}_{2N_t}^C & \cdots & \sum_{j \in \mathcal{N}_t / \{N_t\}} \mathbf{F}_{N_t j}^C \end{bmatrix}, \text{ and}$$

$$\mathbf{F}_{kj}^C = (s_k^j + s_j^k) (\sigma_{n_{kj}}^2)^{-1} \mathbf{h}_{kj}^T \mathbf{h}_{kj}.$$

Based on [25], $\sigma_{n_{kj}}^2 \approx (\Lambda_{kj})^{-1}$ and $\mathbf{h}_{kj}^T = \mathbf{G}_{kj}$. Thus, we can obtain

$$\text{cov}(\Delta \mathbf{x}) \approx (\mathbf{J}_x^A + \mathbf{J}_x^C)^{-1}. \quad (52)$$

This completes the proof that the estimation error of the proposed closed-form solution is in a vicinity of the CRLB.

5. Simulation results

In this section, we conduct Monte-Carlo simulations to verify the proposed node selection approach and CL method. Consider two simulation scenarios, including a 2D environment with 20 agents and a 3D environment with 30 agents. In the 2D environment, the anchors are placed at (0,0), (0,50), (50,0), (50,50), (25,12.5), (12.5,25), (37.5,25), and (25, 37.5). In the 3D environment, the anchors are placed at (0,0,0), (0,50,50), (12.5,0,50), (12.5,50,0), (25,0,0), (25,50,50), (37.5,0,50), (37.5,50,0), (50,0,0), and (50,50,50). The agents are randomly distributed in target area. P and Q are equal to 3 and 4, respectively. The antenna element separation d_a sets as $\lambda/2$. In addition, we assume the normalization condition, i.e., $\int |s(t)|^2 dt = 1$.

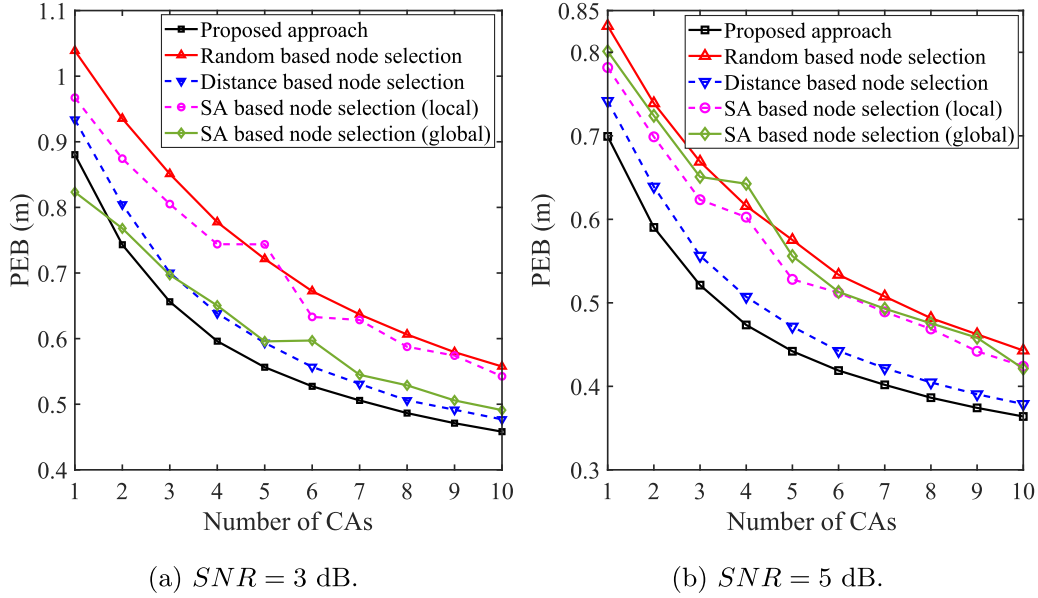


Fig. 3. PEBs for different node selection approaches in 3D environment.

In the AOA CL, set $\varsigma_c = 10^{-4}$ (convergence threshold), and $n_{\text{iter}} = 100$ (maximum number of iterations). The Normalized Root Mean Square Error (NRMSE) is used to evaluate the localization accuracy, $NRMSE = \sqrt{\sum_{j=1}^{M_c} \frac{\|\hat{\mathbf{x}}_u - \mathbf{x}\|}{N_t M_c}}$. $M_c = 1000$ denotes the number of ensemble runs for each simulation.

5.1. PEB analysis

5.1.1. Impact of the number of CAs on PEB

Figure 1 illustrates the average PEBs for different number of CAs (K), varying the SNR value. First, we observe that the cooperation with other agents can improve the localization accuracy significantly. Secondly, the proposed node selection approach can reduce the number of CAs participating in the localization, without significantly affecting accuracy. For example, when $SNR = 5$ dB and $K = 10$ in 2D environment, the PEB of proposed approach is 0.68m with only 0.06m reduction compared with the cooperation with all other agents ($K = 19$). It has 0.92m improvement in PEB compared with localization without cooperation. Third, the performance improvement asymptotically converges to zero with the increase of the number of CAs. Thus, the proposed node selection approach can improve the efficiency of CL by iteratively catching the agents that make the biggest contribution and discarding the least ones. Finally, the larger SNR results in the lower PEB as expected.

5.1.2. Comparison with other node selection approaches

Then, we compare the performance of the proposed approach with other methods, in terms of PEB and time overhead. In the node selection strategy based on distance, the CAs with smaller distance can be selected. In the node selection strategy based on random, the CAs are obtained by random selection in searching region. Note that the average PEB for 100 random results is used to evaluate the performance. In the node selection strategy based on Simulated Annealing (SA), we first use all the candidate CAs' locations to construct the solution space for node selection problem. Then, we adopt two objective functions based SA algorithm to select CAs for the sake of approaching lowest PEB. One is the local optimal problem, defined by (53), where the CA cluster of single agent is obtained at a time. The other is the global optimal prob-

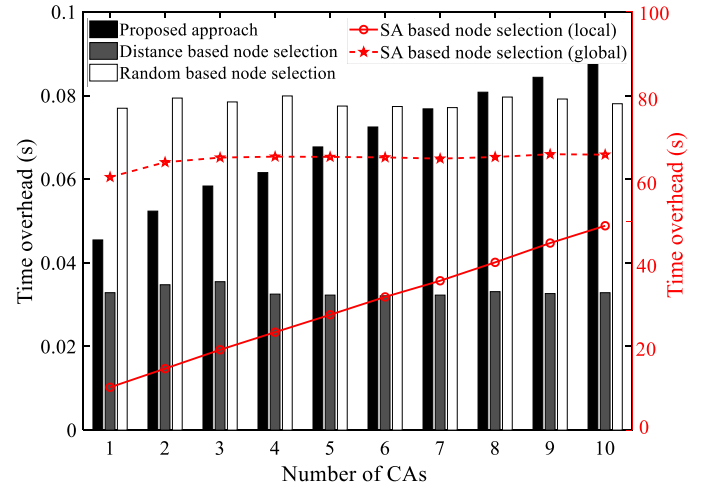


Fig. 4. Time overhead for different node selection approaches.

lem, defined by (54), where the CA clusters of all agents are simultaneously obtained.

$$f_{SA}^1 = \max_{s_k} \text{tr} \left(\mathbf{I} + (\mathbf{J}_k^A)^{-1} \mathbf{J}_k^C \right)^{-1} (\mathbf{J}_k^A)^{-1} \mathbf{J}_k^C (\mathbf{J}_k^A)^{-1}, \text{ for } k = 1, 2, \dots, N_t, \quad (53)$$

where $\mathbf{J}_k^C = \sum_{j \in N_t / \{k\}} \mathbf{C}_{kj}^* - \sum_{j \in N_t / \{k\}} \mathbf{C}_{kj}^* (\mathbf{J}_j^A + \mathbf{C}_{kj}^*)^{-1} \mathbf{C}_{kj}^*$, $\mathbf{C}_{kj}^* = \mathbf{s}_k^j \mathbf{G}_{kj} \mathbf{A}_{kj} \mathbf{G}_{kj}^T$.

$$f_{SA}^2 = \max_{\{s_1, \dots, s_{N_t}\}} \text{tr} \left(\mathbf{I} + (\mathbf{J}_x^A)^{-1} \mathbf{J}_x^C \right)^{-1} (\mathbf{J}_x^A)^{-1} \mathbf{J}_x^C (\mathbf{J}_x^A)^{-1}. \quad (54)$$

Figures 2, 3 show the PEBs for different node selection approaches, varying the number of CAs. It is evident that the proposed scheme can provide lower PEB compared with other methods. When $SNR = 3$ dB and $K = 5$ in the 2D environment, the PEB of proposed scheme is 1.05m, which is smaller than the node selection methods based on random (1.40m), distance (1.15m), SA by f_{SA}^1 (1.18m), and SA by f_{SA}^2 (1.25m).

Figure 4 shows the time overhead for different node selection approaches as K from 1 to 10 in 2D environment. we can find

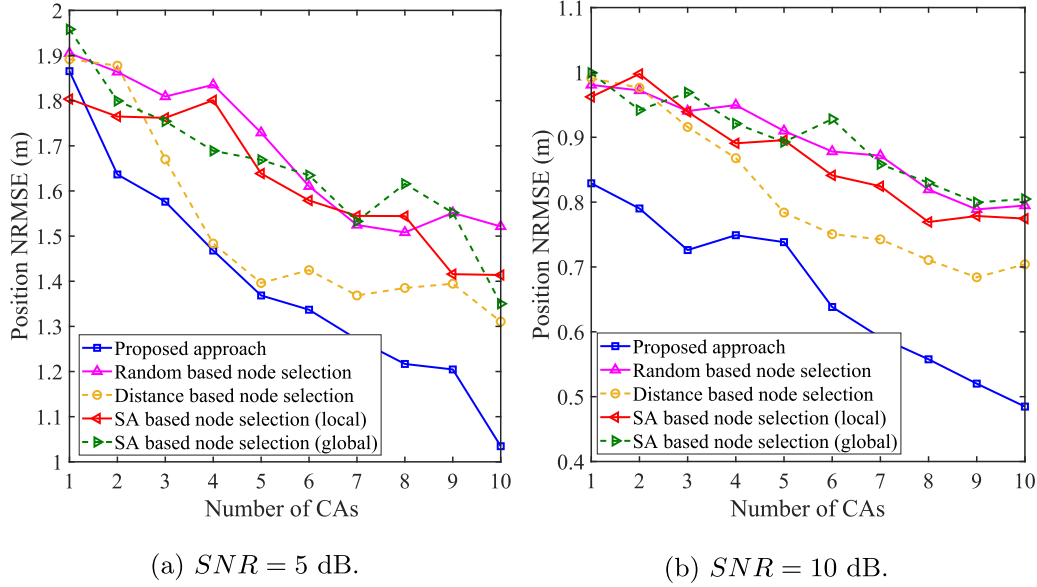


Fig. 5. NRMSEs for different node selection approaches in 2D environment.

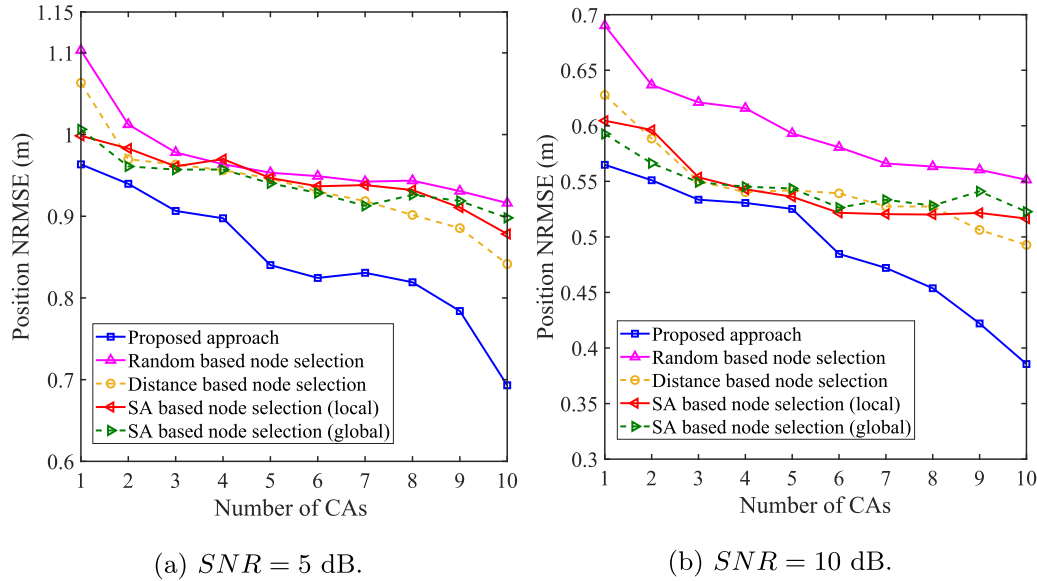


Fig. 6. NRMSEs for different node selection approaches in 3D environment.

that the time overhead by the proposed approach is significantly lower than SA based node selection method, slightly higher than the node selection approaches based on distance and random.

5.2. NRMSE analysis

5.2.1. Comparison with other node selection approaches

Next, we compare the position NRMSEs of different node selection approaches, in which the locations of agents are estimated by the proposed CL method, as shown in Figs. 5, 6. We can find that the localization accuracy of the proposed approach is higher than that of other methods. Furthermore, when the number of CAs increases, the position NRMSE becomes lower, as expected. However, the position NRMSEs sometimes increase slightly when a new CA enters the cooperative network, due to the position uncertainty of the agent.

5.2.2. Comparison with other localization methods

Figures 7, 8 compare the position NRMSEs of the proposed localization method with the LS, 2 phases LS, and iterative LS based localization methods, respectively. It is obvious that the position NRSME of the proposed localization scheme is the lowest among the schemes. When $SNR = 5$ dB and $K = 10$, NRSME of the proposed approach is 1.04m, which has a 0.78m reduction compared with NRSME with no cooperation (1.81m).

5.2.3. Analysis of the convergence probability and iteration number

Tables 1, 2 show the convergence probability and iteration number of the proposed localization method, varying the number of CAs, respectively. The convergence probability is all above 90% and it proves that the proposed method is valid. Moreover, the increase of SNR value will increase the convergence probability, as well as reduce the iteration number.

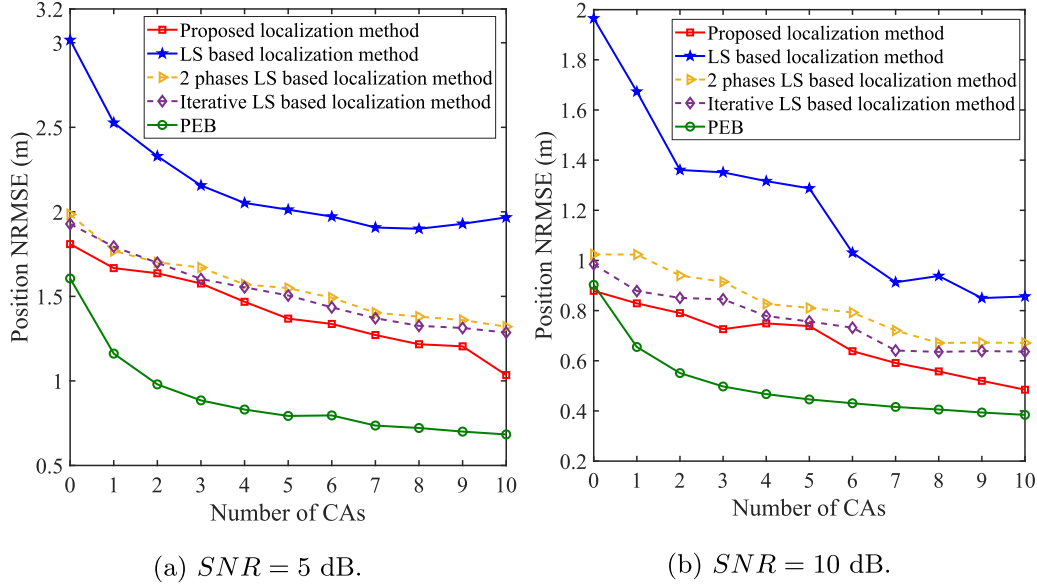


Fig. 7. NRMSEs for different localization methods in 2D environment.

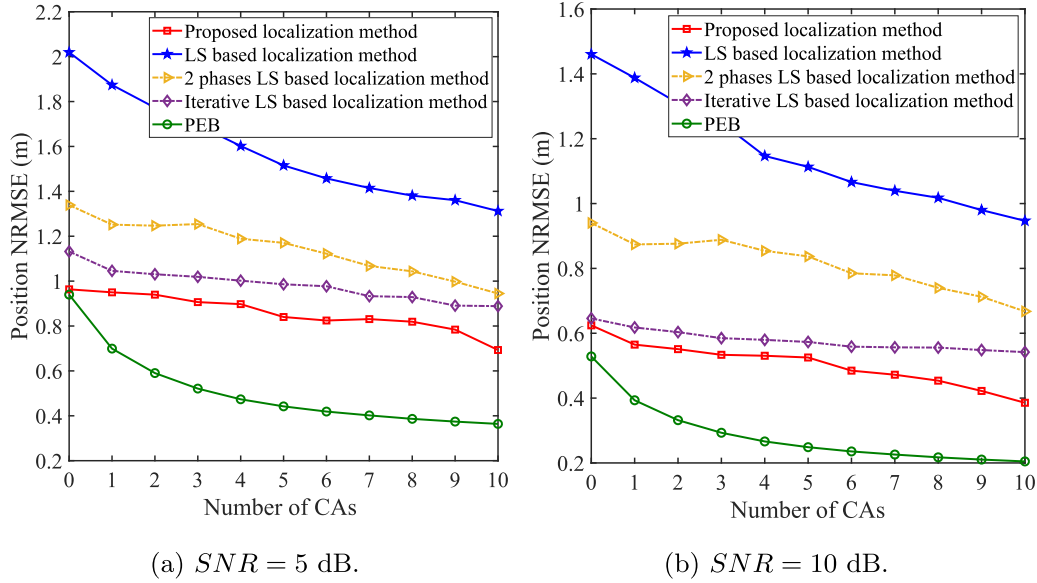


Fig. 8. NRMSEs for different localization methods in 3D environment.

Table 1
Convergence probability (%).

K	1	2	3	4	5	6	7	8	9	10
$SNR = 5$ dB	94	96	93	92	94	90	91	91	91	93
$SNR = 10$ dB	98	99	100	99	100	97	99	100	100	100

Table 2
Iteration number.

K	1	2	3	4	5	6	7	8	9	10
$SNR = 5$ dB	12	15	16	16	16	18	19	19	20	20
$SNR = 10$ dB	8	10	12	13	14	14	14	14	14	15

6. Conclusion

In this paper, we consider a cooperative scenario for simultaneous localization of multiple agents where the AOA measurements between agents and anchors as well as the agent-to-agent mea-

surements are used. In order to alleviate the high complexity and network traffic, the proposed node selection approach has utilized the derived PEB to construct the objective function for the sake of selecting the CAs to minimize the positioning errors. To further improve the localization accuracy, the efficient closed-form solution for AOA based CL has been developed, and an iterative constrained LS method has been proposed to refine the estimations. Numerical simulations have verified the great decrease in number of the agents participating in CL without significant reduction accuracy. Besides, the proposed CL method has better localization accuracy than the existing ones. For the future work, the NLOS signal propagation between nodes will be considered.

Declaration of Competing Interest

Authors declare that there is no conflict of interest in this work.

CRediT authorship contribution statement

Feng Qiu: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Weidong Zhang:** Writing – review & editing, Supervision, Funding acquisition.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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Appendix A

We first analyze the 2D CL, followed by the 3D CL. In order to calculate the FIM respect to the anchors, $\mathbf{J}_x^A$, consider a mapping from $\mathbf{x}$ into another parameter vector, $\boldsymbol{\eta}_A = [\eta_1^A, \dots, \eta_{N_t}^A]^T$, where $\eta_k^A = [\Phi_{k1}, \dots, \Phi_{kN_a}]^T$, $\Phi_{kj} = \varphi_{kj}$ for $j \in \mathcal{N}_a$. Using chain rules, we can have the relationship that $\mathbf{J}_x^A$ for $\mathbf{x}$ is the function of $\mathbf{J}_{\eta_A}^A$ for $\boldsymbol{\eta}_A$,

$$\mathbf{J}_x^A = \mathbf{T}_{\eta_A} \mathbf{J}_{\eta_A}^A (\mathbf{T}_{\eta_A})^T, \quad (\text{A.1})$$

where $\mathbf{T}_{\eta_A}$ denotes the Jacobian matrix for the transformation $\mathbf{x}$ to $\boldsymbol{\eta}_A$, and is

$$\mathbf{T}_{\eta_A} = \frac{\partial \boldsymbol{\eta}_A}{\partial \mathbf{x}} = \text{diag}(\mathbf{G}_1^A, \mathbf{G}_2^A, \dots, \mathbf{G}_{N_t}^A), \quad (\text{A.2})$$

where $\mathbf{G}_k^A = [\mathbf{G}_{k1}, \mathbf{G}_{k2}, \dots, \mathbf{G}_{kN_a}]^T$. Since the received waveforms from different nodes are independent and the complex-valued received signal has a circular Gaussian distribution, by using Slepian Bangs formula, $\mathbf{J}_{\eta_A}^A$ can be written as

$$\mathbf{J}_{\eta_A}^A = \text{diag}(\boldsymbol{\Lambda}_1^A, \boldsymbol{\Lambda}_2^A, \dots, \boldsymbol{\Lambda}_{N_t}^A), \quad (\text{A.3})$$

where $\boldsymbol{\Lambda}_k^A = \text{diag}(\Lambda_{k1}, \dots, \Lambda_{kN_a})$. $\mathbf{G}_{kj}$ and Λ_{kj} are given by (13). Thus, we finally obtain $\mathbf{J}_x^A$ as (12).

Then, we calculate the FIM respect to the cooperation, $\mathbf{J}_x^C$. Similarly, the mapping of $\mathbf{x}$ into $\boldsymbol{\eta}_C = \mathbf{S} \circ [\eta_1^C, \dots, \eta_{N_t}^C]^T$, where $\mathbf{S} = [\mathbf{s}_1^T, \mathbf{s}_2^T, \dots, \mathbf{s}_{N_t}^T]^T$ and $\eta_k^C = [\Phi_{k1}, \dots, \Phi_{kN_t}]^T$, is a bijection. Thus, we can have

$$\mathbf{J}_x^C = \mathbf{T}_{\eta_C} \mathbf{J}_{\eta_C}^C \mathbf{T}_{\eta_C}^T, \quad (\text{A.4})$$

where

$$\mathbf{T}_{\eta_C} = \begin{bmatrix} \mathbf{s}_1^T \circ \mathbf{G}_1^C & -\mathbf{s}_2^T \circ \boldsymbol{\psi}_{12} & \cdots & -\mathbf{s}_{N_t}^T \circ \boldsymbol{\psi}_{1N_t} \\ \mathbf{s}_1^T \circ \boldsymbol{\psi}_{21} & \mathbf{s}_2^T \circ \mathbf{G}_2^C & \cdots & -\mathbf{s}_{N_t}^T \circ \boldsymbol{\psi}_{2N_t} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{s}_1^T \circ \boldsymbol{\psi}_{N_t1} & \mathbf{s}_2^T \circ \boldsymbol{\psi}_{N_t2} & \cdots & \mathbf{s}_{N_t}^T \circ \mathbf{G}_{N_t}^C \end{bmatrix}, \quad (\text{A.5})$$

$\mathbf{G}_k^C = [\mathbf{G}_{k1}, \mathbf{G}_{k2}, \dots, \mathbf{G}_{kN_t}]^T$, $\boldsymbol{\psi}_{kj} = [\psi_{kj}^1, \dots, \psi_{kj}^{N_t}]^T$ has all zero elements except for $\psi_{kj}^k = \mathbf{G}_{kj}$ for $j \in \mathcal{N}_t$. $\mathbf{J}_{\eta_C}^C$ is given by

$$\mathbf{J}_{\eta_C}^C = \text{diag}(\boldsymbol{\Lambda}_1^C, \boldsymbol{\Lambda}_2^C, \dots, \boldsymbol{\Lambda}_{N_t}^C), \quad (\text{A.6})$$

where $\boldsymbol{\Lambda}_k^C = \text{diag}(\Lambda_{k1}, \dots, \Lambda_{kN_t})$. Thus, we finally obtain $\mathbf{J}_x^C$ as (15).

For 3D CL, the modification needed is to replace $\Phi_{kj} = \varphi_{kj}$ with $\Phi_{kj} = [\varphi_{kj}, \alpha_{kj}]$. Similarly, we can obtain $\mathbf{J}_x^A$ and $\mathbf{J}_x^C$ in (14) and (15).

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